Varshitha Gogineni

☎ +1 (940) 300-9984
✉ varshithagogineni007@gmail.com ▼ https://www.linkedin.com/in/varshitha-gogineni/

Summary

AI & Data Engineer recognized at Y Combinator's Gemini x Pipecat Hackathon and Dallas AI 2025 (Top 14 of 32 teams), with hands-on experience building cloud-native data pipelines, LLM-powered applications, and real-time analytics on Google Cloud Platform. Skilled in Python, SQL, PySpark, BigQuery, Dataproc, and Pub/Sub, with proven expertise in RAG, Agentic AI, Tool Calling, and informatics-driven analytics on healthcare datasets. Adept at integrating LLMs with backend services and BI dashboards to deliver production-ready, measurable solutions.

Education

University of North Texas *May 2026*
Master of Science in Information Science (GPA: 4.00 / 4.00) *Denton, TX*

- **Relevant Coursework:** Natural Language Processing, Information Retrieval, Machine Learning, Data Mining, Database Systems, Software Development for AI

BV Raju Institute of Technology *May 2024*
Bachelor of Technology in Information Technology *Hyderabad, India*

Experience

Cambo Box *Aug 2025 - Dec 2025*
AI Engineer Intern *United States, Remote*

- Designed a production-ready agentic AI voice workflow for restaurants, autonomously handling 70-80% of inbound calls and automating outbound reservation reminders.
- Integrated AI voice agents with Twilio telephony supporting real-time barge-in and seamless escalation, improving call-handling efficiency by ~40%.
- Architected tool-calling pipelines for 100+ reservation scenarios (create, modify, cancel, lookup) with strict JSON schema validation, reducing invalid AI tool calls by ~90%.
- Built REST API orchestration and cron-driven outbound workflows for sub-second response and zero-touch daily reservation confirmations.

University of North Texas - CMHT IT Services *Dec 2024 - May 2026*
IT Services Team Lead & Web Developer *Denton, TX*

- Promoted from classroom technology assistant to Team Lead, managing a team of part-time student assistants - running weekly syncs and coordinating work on 180+ managed laptops for students and CMHT faculty.
- Develop API-driven web applications and internal data tooling supporting 5+ academic departments, reducing average response time on key endpoints by ~30%.
- Built ETL utilities and n8n productivity automations to auto-generate reports and route form submissions, eliminating ~10 hours of manual staff work per week and cutting report turnaround from hours to under 5 minutes.
- Maintained team SOPs and IT documentation for accuracy and clarity, while providing hardware/software troubleshooting and end-user support across CMHT.

Projects

Healthcare Data Lifecycle - Cloud-Native Pipeline (GCP) | *BigQuery, PySpark, Hive, Dataproc, Pub/Sub, Power BI*

- Built an end-to-end GCP data pipeline processing 1,048,575 CDC BRFSS healthcare survey records (2020-2024) into real-time informatics insights.
- Engineered scalable ETL, batch, and streaming workflows with Cloud Storage, BigQuery, and Apache Spark + Hive on Dataproc; reduced health-alert delivery latency from minutes to under 5 seconds via Pub/Sub streaming into BigQuery.
- Delivered Power BI dashboards on depression trends, income-based health disparities, and behavioral risk patterns, enabling near real-time public health monitoring across 27 BRFSS variables.

CMHT Voice Agent | *Gemini APIs, LangChain, Python, RAG*

- Built a real-time LLM voice assistant with Gemini and LangChain, modular tool-calling, and session memory for context-aware multi-turn workflows.
- Optimized API latency and token usage via prompt caching, batched requests, and efficient context window management.

VoiceFit - AI Voice Assistant | *Whisper STT, React, n8n, OpenAI*

- Built a voice-enabled AI assistant with Whisper STT and a responsive React front-end; integrated speech-to-text with backend logic for hands-free task execution.
- Designed automated task workflows in n8n reducing manual interaction steps by 60%+ for routine operations.

NYC Taxi Trips Data Engineering Pipeline | *BigQuery*, *Apache Airflow*, *dbt*, *Great Expectations*, *Python*

- Built an end-to-end batch data pipeline ingesting 10M+ NYC Yellow Taxi trips into BigQuery with bronze/silver/gold medallion layering; modeled curated analytics tables in dbt to surface daily trip volume, fare trends, and hourly demand patterns.
- Orchestrated ingestion, transformations, and Great Expectations data quality checks (null, range, uniqueness, schema validation) as Apache Airflow DAGs with retries and failure alerts; published metrics in a Looker Studio dashboard for at-a-glance KPI monitoring.

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript, PySpark, HiveQL
Data Engineering & Big Data: Apache Spark, Hive, Dataproc, BigQuery, Apache Airflow, dbt, Great Expectations, Cloud Storage, Pub/Sub, ETL/ELT, Power BI, Looker Studio
AI & Automation: RAG, Agentic AI, Tool Calling, LangChain, Gemini & OpenAI APIs, Whisper STT, NLP, CNNs, n8n, Model Context Protocol (MCP)
Cloud & Tools: GCP, Azure AI, Docker, Twilio, Git/GitHub
Databases: PostgreSQL, MongoDB, MySQL, Firebase, Pinecone Vector DB

Certifications

- **Anthropic -** Introduction to Agent Skills, Introduction to Model Context Protocol (MCP), Building with the Claude API, Claude Code in Action (Apr 2026)

Achievements

- Dallas AI 2025 - Top 14 of 32 teams for VoiceFit, an AI-powered hands-free fitness companion (OpenAI Whisper, n8n, Supabase); live at voicefit.vercel.app.
- Participated in the Google Gemini x Pipecat Hackathon at Y Combinator (SF Tech Week 2025) - co-built an AI-powered Mock Interview Platform with Gemini Live and Pipecat enabling real-time voice, screen-share, and AI code review.
- Built CMHT Voice Assistant for UNT's College of Merchandising, Hospitality and Tourism - a speech-to-speech AI agent that helps students and faculty locate room numbers and provides front-desk assistance.